

# Enterprise Network Design & Implementation

Switching & Routing Essentials • Cisco Packet Tracer • Academic Portfolio

|       |      |                 |
|-------|------|-----------------|
| VLANs | OSPF | WLAN & Security |
|-------|------|-----------------|

# Project Overview

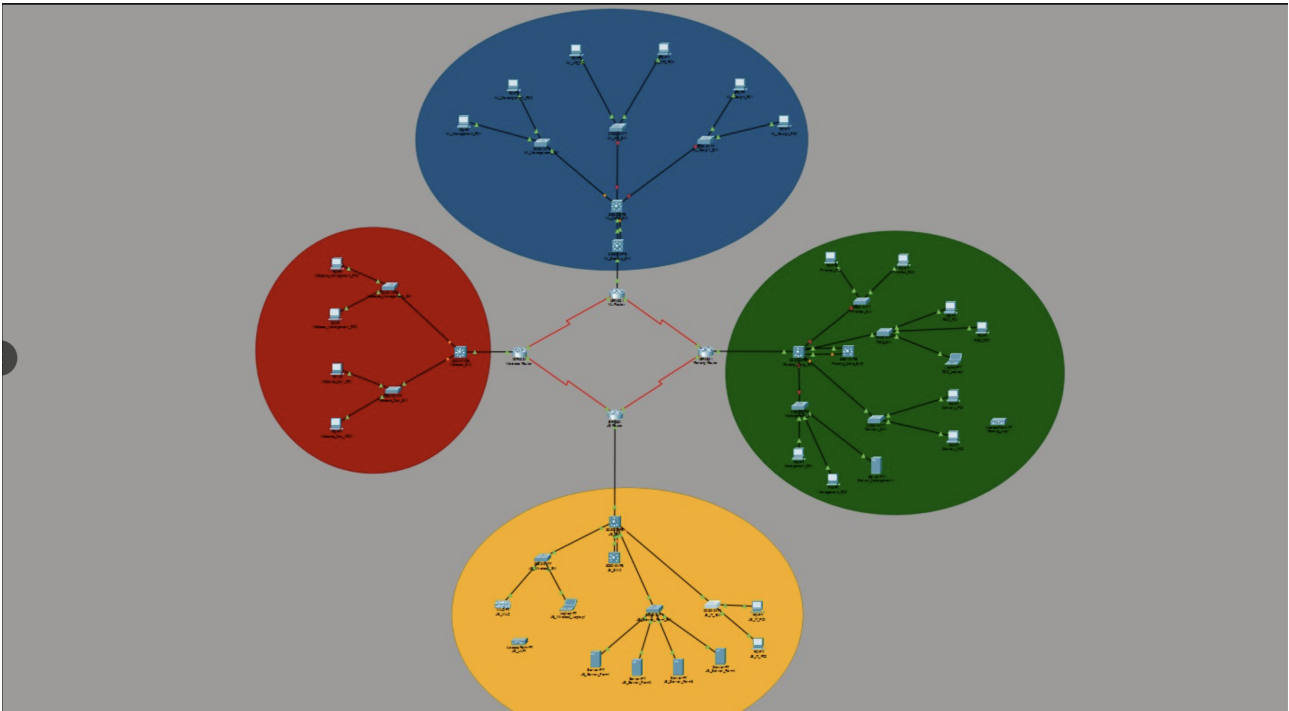
This academic project designed and implemented a multi-branch enterprise network prototype for Shelloos using Cisco Packet Tracer. The design covered Kuala Lumpur (HQ), Malacca, Penang and Johor Bahru, with WAN connectivity between branches and a centralized server farm.

## Core technologies

- VLAN segmentation and inter-VLAN routing
- VLSM-based IP addressing
- OSPF and default static routing
- HSRP redundancy and EtherChannel
- WLAN/WLC and WPA2 security
- DHCP, DNS, FTP and Web services
- Layer 2 security controls including Port Security and BPDU Guard
- Connectivity, routing and service verification

The original report describes the objective as developing an enterprise network solution combining switching, routing, wireless networking, redundancy and Layer 2 security mechanisms.

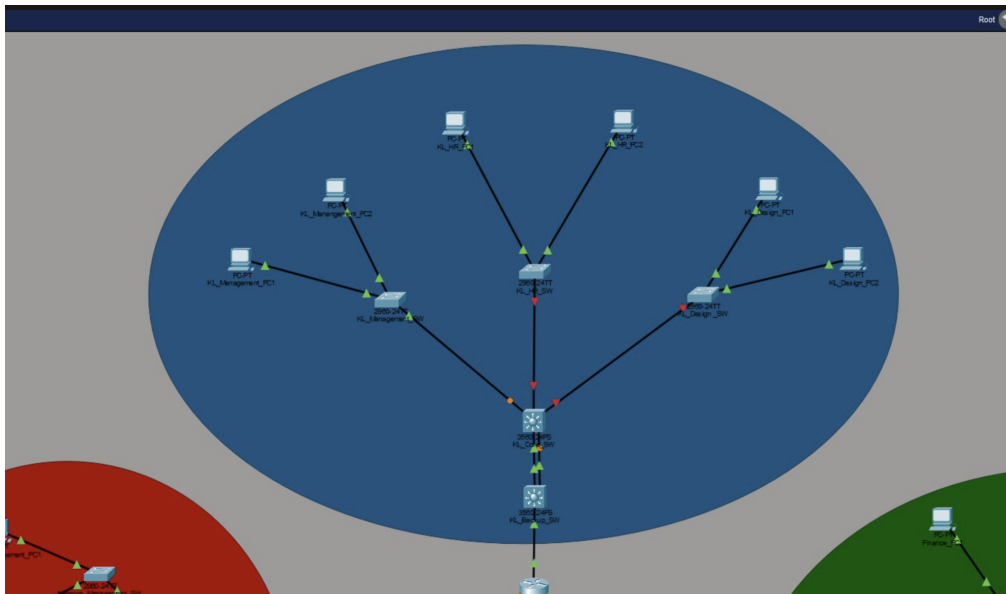
# Enterprise Topology



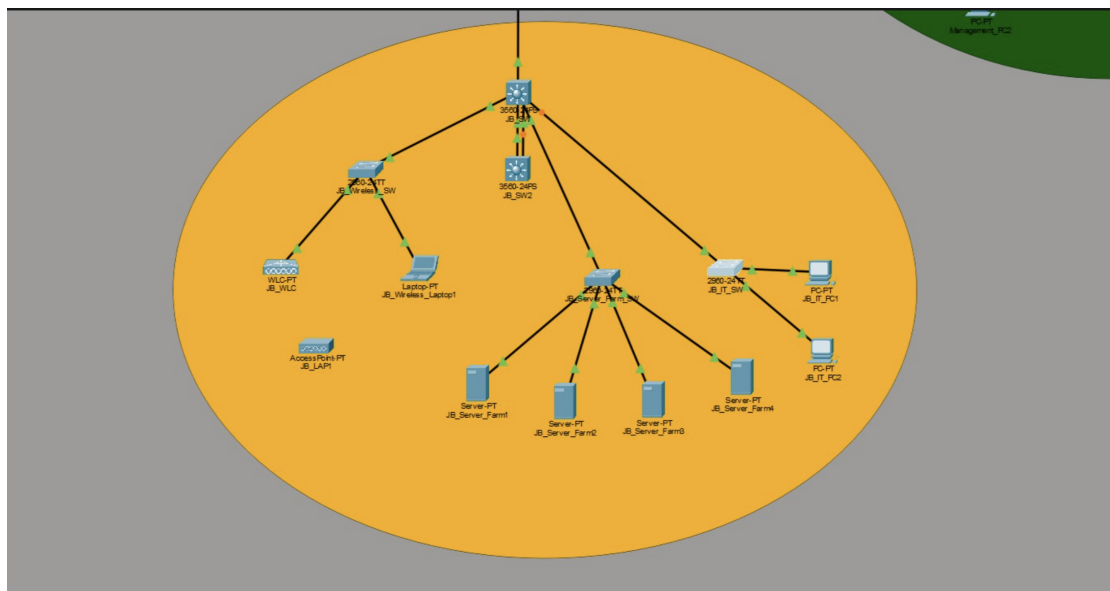
Selected screenshot: overall enterprise topology showing the four-branch architecture.

The topology connects four locations through WAN links and uses VLAN segmentation to separate departmental traffic. The Johor Bahru branch also hosts centralized DNS, FTP, Web and DHCP services.

# Branch Network Design



Selected screenshot: Kuala Lumpur HQ topology with departmental VLAN segmentation and redundant core design.



Selected screenshot: Johor Bahru branch topology and centralized services environment.

# Configuration & Implementation

The implementation included router and switch configuration, VLANs, trunking, inter-VLAN routing, WAN routing, redundancy and security controls. The original report documents the use of Cisco IOS-style configuration commands and Packet Tracer interfaces to implement these features.

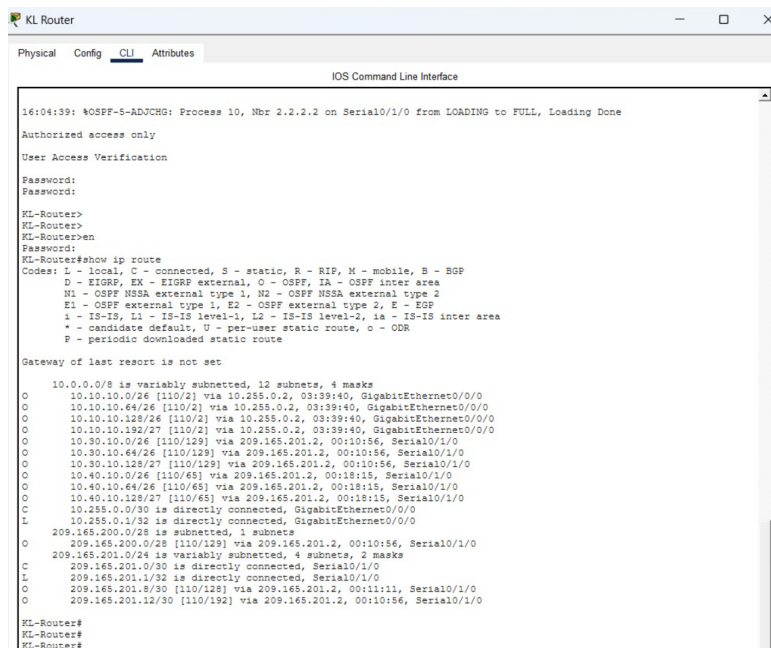
```
interface range f0/11-12
  description EtherChannel_to_Penang_Core_SW2
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 110,120,130,140,199,999
  channel-group 2 mode active
  no shutdown
interface port-channel 2
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 110,120,130,140,199,999
interface f0/21
  description trunk_to_Management_SW
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 140,999
```

Selected configuration screenshot from the original Packet Tracer implementation.

```
no shutdown
interface range f0/23-24
  description EtherChannel_to_JB_SW2
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 200,210,220,299,999
  channel-group 3 mode active
  no shutdown
interface port-channel 3
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 200,210,220,299,999
```

Selected routing/configuration screenshot showing implementation work.

# Routing & Network Services



```
KL Router
Physical Config CLI Attributes
IOS Command Line Interface

16:04:39: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on Serial0/1/0 from LOADING to FULL, Loading Done

Authorized access only
User Access Verification
Password:
Password:

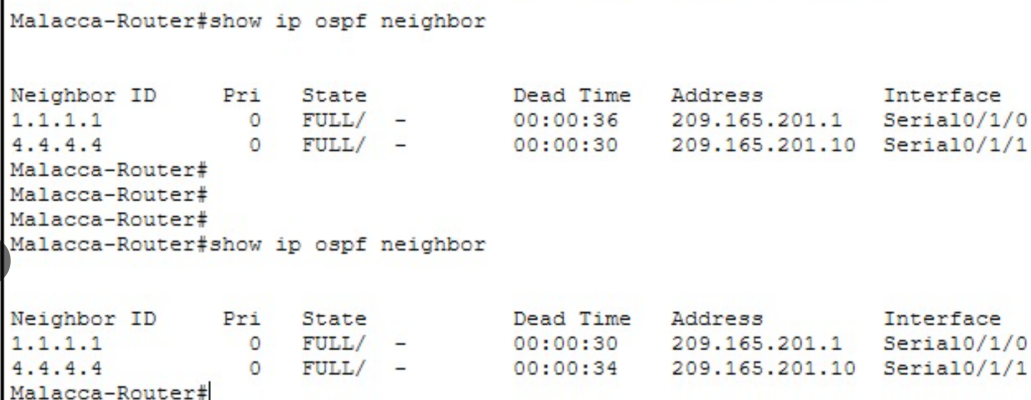
KL-Router>
KL-Router>
KL-Router>show ip route
KL-Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       O - OSPF, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 12 subnets, 4 masks
O 10.10.10.0/26 [110/2] via 10.255.0.2, 03:39:40, GigabitEthernet0/0/0
O 10.10.10.64/26 [110/2] via 10.255.0.2, 03:39:40, GigabitEthernet0/0/0
O 10.10.10.128/26 [110/2] via 10.255.0.2, 03:39:40, GigabitEthernet0/0/0
O 10.10.10.192/27 [110/2] via 10.255.0.2, 03:39:40, GigabitEthernet0/0/0
O 10.30.10.0/26 [110/129] via 209.165.201.2, 00:10:56, Serial0/1/0
O 10.30.10.64/26 [110/129] via 209.165.201.2, 00:10:56, Serial0/1/0
O 10.30.10.128/27 [110/129] via 209.165.201.2, 00:10:56, Serial0/1/0
O 10.40.10.0/26 [110/65] via 209.165.201.2, 00:18:15, Serial0/1/0
O 10.40.10.64/26 [110/65] via 209.165.201.2, 00:18:15, Serial0/1/0
O 10.40.10.128/27 [110/65] via 209.165.201.2, 00:18:15, Serial0/1/0
C 10.255.0.0/30 is directly connected, GigabitEthernet0/0/0
L 10.255.0.1/32 is directly connected, GigabitEthernet0/0/0
209.165.200.0/28 is subnetted, 1 subnets
O 209.165.200.0/28 [110/129] via 209.165.201.2, 00:10:56, Serial0/1/0
209.165.201.0/24 is variably subnetted, 4 subnets, 2 masks
C 209.165.201.0/30 is directly connected, Serial0/1/0
L 209.165.201.1/32 is directly connected, Serial0/1/0
O 209.165.201.8/30 [110/128] via 209.165.201.2, 00:11:11, Serial0/1/0
O 209.165.201.12/30 [110/192] via 209.165.201.2, 00:10:56, Serial0/1/0

KL-Router#
KL-Router#
KL-Router#
```

Selected screenshot showing routing configuration/output.



```
Malacca-Router#show ip ospf neighbor

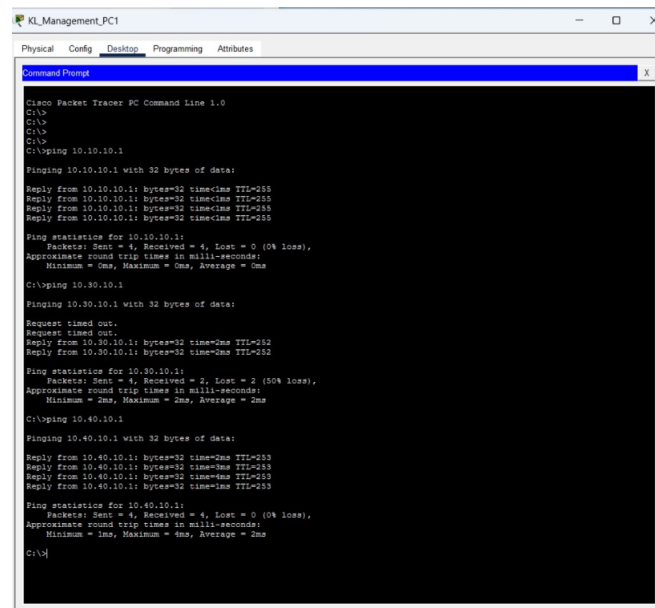
Neighbor ID    Pri   State           Dead Time   Address        Interface
1.1.1.1         0    FULL/-          00:00:36    209.165.201.1  Serial0/1/0
4.4.4.4         0    FULL/-          00:00:30    209.165.201.10 Serial0/1/1
Malacca-Router#
Malacca-Router#
Malacca-Router#
Malacca-Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
1.1.1.1         0    FULL/-          00:00:30    209.165.201.1  Serial0/1/0
4.4.4.4         0    FULL/-          00:00:34    209.165.201.10 Serial0/1/1
Malacca-Router#
```

Selected verification screenshot from the network implementation.

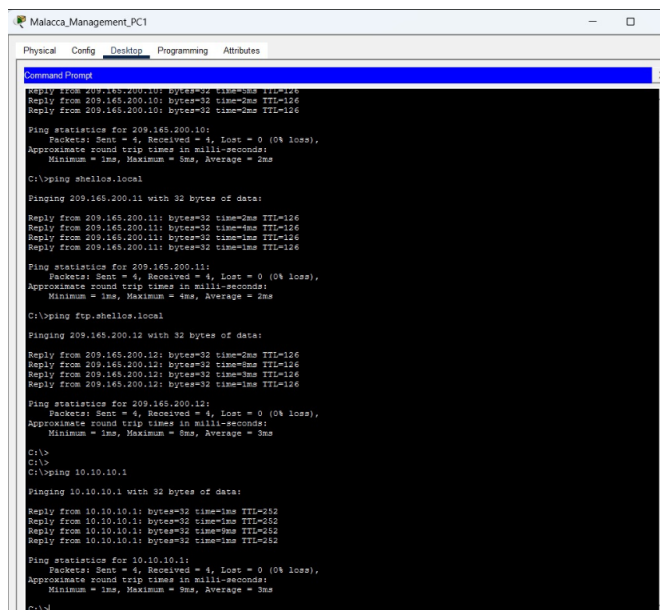
The project used OSPF for dynamic routing between branch networks, alongside a default static route design where appropriate. Centralized server services were also configured for DHCP, DNS, FTP and Web access.

# Testing & Verification



```
KIL_Management_PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>
C:\>
C:\>
C:\>ping 10.10.10.1
Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time=1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=1ms TTL=255
Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.30.10.1
Pinging 10.30.10.1 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.30.10.1: bytes=32 time=2ms TTL=252
Reply from 10.30.10.1: bytes=32 time=2ms TTL=252
Ping statistics for 10.30.10.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\>ping 10.40.10.1
Pinging 10.40.10.1 with 32 bytes of data:
Reply from 10.40.10.1: bytes=32 time=2ms TTL=253
Reply from 10.40.10.1: bytes=32 time=2ms TTL=253
Reply from 10.40.10.1: bytes=32 time=2ms TTL=253
Reply from 10.40.10.1: bytes=32 time=2ms TTL=253
Ping statistics for 10.40.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 2ms
C:\>
```

Selected screenshot: command-line verification output from the completed network.



```
Malacca_Management_PC1
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 209.165.200.10: bytes=32 time=2ms TTL=126
Reply from 209.165.200.10: bytes=32 time=2ms TTL=126
Reply from 209.165.200.10: bytes=32 time=2ms TTL=126
Ping statistics for 209.165.200.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 2ms
C:\>ping shellcs.local
Pinging 209.165.200.11 with 32 bytes of data:
Reply from 209.165.200.11: bytes=32 time=2ms TTL=126
Reply from 209.165.200.11: bytes=32 time=2ms TTL=126
Reply from 209.165.200.11: bytes=32 time=2ms TTL=126
Reply from 209.165.200.11: bytes=32 time=2ms TTL=126
Ping statistics for 209.165.200.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 2ms
C:\>ping ftp.shellcs.local
Pinging 209.165.200.12 with 32 bytes of data:
Reply from 209.165.200.12: bytes=32 time=2ms TTL=126
Reply from 209.165.200.12: bytes=32 time=2ms TTL=126
Reply from 209.165.200.12: bytes=32 time=2ms TTL=126
Reply from 209.165.200.12: bytes=32 time=2ms TTL=126
Ping statistics for 209.165.200.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 2ms
C:\>
C:\>ping 10.10.10.1
Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time=1ms TTL=252
Reply from 10.10.10.1: bytes=32 time=1ms TTL=252
Reply from 10.10.10.1: bytes=32 time=1ms TTL=252
Reply from 10.10.10.1: bytes=32 time=1ms TTL=252
Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 2ms
C:\>
```

Selected screenshot: network verification output used to confirm configuration behaviour.

The report includes end-to-end connectivity testing, inter-site connectivity, DNS/name resolution, FTP service verification, gateway reachability and OSPF neighbour verification.

# Security & Key Learning Outcomes

- Understanding enterprise network topology design across multiple sites
- Applying VLANs and inter-VLAN routing for logical segmentation
- Implementing OSPF and validating routing behaviour
- Understanding redundancy with HSRP and EtherChannel
- Configuring wireless networking and WPA2 authentication
- Applying Layer 2 security controls such as Port Security and BPDU Guard
- Testing network connectivity and interpreting command-line verification output

## Portfolio Summary

This project strengthened my practical understanding of enterprise networking, switching, routing, wireless infrastructure and network security using Cisco Packet Tracer.

**Suggested LinkedIn skills:** Cisco Packet Tracer • Networking • Switching • Routing • VLAN • OSPF • EtherChannel • HSRP • WLAN • Network Security • DHCP • DNS